QuantumDynamics Framework and Capability Notes

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Abstract

QuantumDynamics is a family of one-, two-, and three-coordinate, two-diabatic-state wavepacket and reduced-density-matrix tools for hydrogen-atom transfer. It combines split-operator propagation, optional Markovian or exponentially correlated Langevin centroid damping, optional stochastic modulation of either well, MPI-distributed trajectories, OpenMP/FFTW acceleration, and physical-unit input/output. It also provides quantum-detailed-balance density dynamics, strong-coupling non-Markovian HEOM, an exact finite spin-bath model, MPI-IO single-file output, exact restart, convergence campaigns, and controlled mechanism attribution. This note records equations, controls, tested behavior, analysis capability, and method boundaries.

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1 Scope and current status

Current code supports reproducible pure-state trajectories, ensemble kinetics, two- and three-coordinate wavepacket models, and finite-grid reduced density matrices under weak-coupling Davies or strong-coupling HEOM dynamics. Input validation, unit conversion, FFT ordering, random-stream portability, restart identity, serial and parallel HDF5, state-selective damping, and output safety have regression coverage. It remains a parameterized nuclear-dynamics model, not a first-principles electronic-structure engine.

The available bath descriptions form a hierarchy rather than interchangeable implementations. Closed split-operator propagation isolates coherent wavepacket motion. Centroid Langevin dynamics adds a classical thermalization model to pure states. Davies dynamics gives a weak-coupling quantum Markov semigroup with exact detailed balance. HEOM resolves finite Gaussian bath memory nonperturbatively, while the finite spin bath supplies an exact non-Gaussian reference with recurrences. A scientific calculation should select the least elaborate level that contains the relevant time scales and then test its regime assumptions; agreement between two levels is stronger evidence than numerical convergence inside only one model.

2 Dynamical model

2.1 Two-state Hamiltonian

Wavefunction is

$$\Psi(x,t) = \begin{pmatrix} \psi_1(x,t) \\ \psi_2(x,t) \end{pmatrix}, \tag{1}$$

with Hamiltonian

$$\hat{H} = -\frac{1}{2m}\frac{\partial^2}{\partial x^2}\mathbb{1}_{2\times 2} + \begin{pmatrix} V_{11}(x,t) & V_{12}(x) \\ V_{12}(x) & V_{22}(x,t) \end{pmatrix}. \tag{2}$$

Base diabatic wells are

$$V_{11}(x) = \tfrac{1}{2}k_1(x-x_1)^2 + c_{4,1}(x-x_1)^4 + \Delta_1, \tag{3}$$

$$V_{22}(x) = \tfrac{1}{2}k_2(x-x_2)^2 + c_{4,2}(x-x_2)^4 + \Delta_2. \tag{4}$$

Quartic terms are enabled by anharmonic model selections. Gaussian coupling is

$$V_{12}(x) = V_c \exp\left[-\frac{(x-x_c)^2}{2\sigma_c^2}\right], \tag{5}$$

with an exponential alternative. Lower and upper adiabatic surfaces are the eigenvalues of the local 2×2 diabatic potential matrix,

$$V_{\pm}(x) = \frac{V_{11}+V_{22}}{2} \pm \sqrt{\left(\frac{V_{11}-V_{22}}{2}\right)^2 + |V_{12}|^2}. \tag{6}$$

At a diabatic crossing $V_{11} = V_{22}$, nonzero coupling opens an adiabatic gap $2|V_{12}|$. For real coupling, the local mixing angle may be chosen through $\tan 2\theta = 2V_{12}/(V_{11}-V_{22})$. This makes clear why "reactant" and "product" populations are representation-dependent observables: the code reports diabatic populations

$$P_i(t) = \int |\psi_i(x,t)|^2\, dx, \qquad P_1 + P_2 = \|\Psi\|^2, \tag{7}$$

whereas adiabatic populations require the coordinate-dependent rotation by $\theta(x)$. Diabatic representation avoids explicit derivative couplings, but a first-principles diabatic matrix must retain a consistent electronic phase and state character across geometry.

2.2 Split propagation

One Hamiltonian step uses symmetric potential splitting,

$$e^{-i\hat{H}\Delta t} \approx e^{-i\hat{V}\Delta t/2} e^{-i\hat{T}\Delta t} e^{-i\hat{V}\Delta t/2}. \tag{8}$$

Local potential exponentials use an analytic Hermitian 2×2 form. Kinetic propagation uses forward and inverse FFTW transforms. Dedicated plans are created for both diabatic components. Even and odd grid sizes have tested FFT frequency ordering.

This symmetric, time-reversible Strang factorization has local error $O(\Delta t^3)$ and accumulated fixed-time error $O(\Delta t^2)$ when the commutators of $\hat{T}$ and $\hat{V}$ are sufficiently regular. Each substep is unitary, so the discrete norm is conserved to roundoff when damping and absorption are disabled. In Fourier space,

$$\hat{T} \longrightarrow \frac{k^2}{2m}, \qquad k_n = \frac{2\pi n}{L}, \tag{9}$$

and kinetic propagation is diagonal. FFT representation also imposes a periodic box and a Nyquist cutoff. Grid-spacing convergence therefore tests unresolved high momenta, while box-size and absorber tests diagnose periodic-image return. The method follows the split spectral construction of Feit, Fleck, and Steiger (JCP 47, 412 (1982)) and the symmetric composition of Strang (SIAM J. Numer. Anal. 5, 506 (1968)).

2.3 Momentum Langevin bath

Markovian damping uses an exact finite-step Ornstein–Uhlenbeck centroid update consistent with classical fluctuation–dissipation statistics. Colored damping uses an exponential memory kernel,

$$\gamma(t) = \frac{\gamma}{\tau_c} e^{-t/\tau_c}, \qquad \tau_c = \frac{2}{\mathrm{FWHM}}, \tag{10}$$

with auxiliary memory and stationary random-force variables. The controls `damp_reactant` and `damp_product` select the diabatic component that receives the momentum translation. Enabling both recovers shared-centroid damping.

For white damping, the free centroid momentum update is the exact Ornstein–Uhlenbeck transition

$$\bar{p}_{n+1} = e^{-\gamma\Delta t}\bar{p}_n + \sqrt{\frac{m}{\beta}\left(1 - e^{-2\gamma\Delta t}\right)}\,\xi_n, \qquad \xi_n \sim \mathcal{N}(0,1), \tag{11}$$

which preserves the Maxwell variance $\langle \bar{p}^2 \rangle = m/\beta$ for any time step. Colored damping represents

$$\dot{\bar{p}}(t) = F(t) - \int_0^t \gamma(t-s)\bar{p}(s)\,ds + R(t), \qquad \langle R(t)R(t')\rangle = \frac{m}{\beta}\gamma(|t-t'|). \tag{12}$$

The exponential kernel admits a Markovian embedding with one memory coordinate and one OU random force, avoiding storage of the full momentum history. This is the classical generalized Langevin FDT described by Kubo (Rep. Prog. Phys. 29, 255 (1966)).

This thermostat is a classical-FDT pure-state construction. The QGLE label does not imply a full quantum-frequency-dependent fluctuation–dissipation kernel. Because the kick acts on a wavepacket centroid, it constrains selected first moments and their thermal variance; it does not by itself generate a completely positive reduced-density evolution or the frequency-dependent zero-point fluctuations of a quantum bath. State-selective damping is therefore a controlled phenomenological model whose sensitivity must be checked by comparing reactant-only, product-only, both-state, and bath-free calculations.

2.4 Quantum-FDT density-matrix path

`quantum_fdt_density.py` builds the full finite-grid system Hamiltonian, diagonalizes it, and propagates the reduced density matrix with a completely positive secular Davies generator. Quantum and classical FDT differ most clearly in frequency space. Up to spectral-density convention, equilibrium quantum noise has

$$S_{\mathrm{sym}}(\omega) \propto J(|\omega|) \coth\left(\frac{\beta|\omega|}{2}\right), \qquad S(-\omega) = e^{-\beta\omega} S(\omega). \tag{13}$$

The first relation includes zero-point fluctuations; the second is the KMS absorption/emission asymmetry. Classical FDT is recovered only when $\beta|\omega| \ll 1$, where $\coth(\beta|\omega|/2) \simeq 2/(\beta|\omega|)$. Thus temperature alone does not determine whether classical noise is adequate: all dynamically important Bohr frequencies must also satisfy the high-temperature condition. See the quantum FDT development of Kubo (Physica 38, 349 (1968)).

If $H_S|n\rangle = E_n|n\rangle$, the bath coupling operator is resolved into Bohr-frequency components of Q. For each positive frequency $\omega = E_n - E_m$, Drude–Ohmic emission and absorption rates are

$$\Gamma_{n\to m} = gJ(\omega)[n_B(\omega)+1]|Q_{mn}|^2, \tag{14}$$

$$\Gamma_{m\to n} = gJ(\omega)n_B(\omega)|Q_{mn}|^2, \tag{15}$$

where $n_B(\omega) = (e^{\beta\omega}-1)^{-1}$ and the implemented dimensionless spectral shape is

$$J(\omega) = \frac{\omega/\omega_c}{1+(\omega/\omega_c)^2}. \tag{16}$$

Thus

$$\frac{\Gamma_{m\to n}}{\Gamma_{n\to m}} = e^{-\beta\omega}. \tag{17}$$

Energy populations obey the Pauli equation

$$\dot{p}_n = \sum_{m\neq n} \left(\Gamma_{m\to n} p_m - \Gamma_{n\to m} p_n\right), \tag{18}$$

while nondegenerate coherences rotate at their Bohr frequencies and decay at one half the sum of outgoing population rates plus zero-frequency pure dephasing. This KMS relation enforces quantum detailed balance at every resolved transition and makes the Gibbs density matrix stationary. Stored output includes the full complex density matrix, physical coordinate densities, populations, energies, and a KMS residual. The solver now supports three explicit preparations. A localized wavepacket supplies a rank-one pure state, while the exact finite-basis canonical mixture uses

$$\rho_\beta = \frac{e^{-\beta H_S}}{Z}, \qquad p_n = \frac{e^{-\beta(E_n-E_0)}}{\sum_m e^{-\beta(E_m-E_0)}}. \tag{19}$$

Subtracting E_0 prevents numerical overflow without changing the normalized weights. A stochastic thermal-pure estimator independently samples complex Gaussian vectors $|r_s\rangle$, Boltzmann filters them,

$$|\phi_s\rangle = e^{-\beta H_S/2}|r_s\rangle, \qquad \widehat{\rho}_\beta = \frac{\sum_s |\phi_s\rangle\langle\phi_s|}{\sum_s \langle\phi_s|\phi_s\rangle}, \tag{20}$$

and reports trace-estimator, Frobenius, population, and effective-sample-size diagnostics. With sample weights $w_s = \langle\phi_s|\phi_s\rangle$, the latter is

$$N_{\mathrm{eff}} = \frac{(\sum_s w_s)^2}{\sum_s w_s^2}. \tag{21}$$

Both thermal preparations are canonical only within the retained finite-grid energy basis; temperature, box, grid, and retained-state convergence remain mandatory. This is the controlled weak-coupling, Markovian

quantum-FDT limit. Strong-coupling non-Markovian quantum baths require HEOM or an equivalent influence-functional method and are outside this solver's declared regime. They are handled by the separate HEOM path below.

Davies construction is controlled by weak system–bath coupling, bath correlation times short relative to relaxation, and secular separation of distinct Bohr frequencies. Secularization guarantees complete positivity but suppresses coupling between populations and coherences at nearly degenerate transitions. Consequently, trace preservation, positivity, and KMS stationarity verify internal consistency; they do not prove that weak-coupling or secular assumptions fit a chosen physical system. This limit follows Davies' Markovian weak-coupling construction (Commun. Math. Phys. 39, 91 (1974)).

2.5 Strong-coupling non-Markovian HEOM

`heom_nonmarkovian.py` treats a harmonic Gaussian environment with linear system coupling operator Q and Drude–Lorentz spectral density

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \tag{22}$$

where λ is reorganization energy and γ^{-1} is bath memory time. The finite-temperature bath correlation is decomposed as

$$C(t) = \sum_{k=0}^{K} c_k e^{-\nu_k t} + 2\Delta_K \delta(t), \tag{23}$$

with one Drude pole, K Matsubara poles, and residual white-tail correction Δ_K. In atomic units, the retained expansion uses

$$\nu_0 = \gamma, \qquad c_0 = \lambda\gamma\left[\cot\left(\frac{\beta\gamma}{2}\right) - i\right], \tag{24}$$

$$\nu_k = \frac{2\pi k}{\beta}, \qquad c_k = \frac{4\lambda\gamma\nu_k}{\beta(\nu_k^2 - \gamma^2)}, \quad k \geq 1, \tag{25}$$

and chooses the residual term so that

$$\sum_{k=0}^{K} \frac{\mathrm{Re}\, c_k}{\nu_k} + \Delta_K = \frac{2\lambda}{\beta\gamma}. \tag{26}$$

This sum rule preserves the exact integrated real bath correlation after truncating the Matsubara series. Scaled auxiliary density operators $\tilde{\rho}_{\boldsymbol{n}}$ obey

$$\begin{aligned} \dot{\tilde{\rho}}_{\boldsymbol{n}} = & -\left(i\mathcal{L} + \sum_k n_k\nu_k + \Delta_K Q^\times Q^\times\right)\tilde{\rho}_{\boldsymbol{n}} \\ & - i\sum_k \sqrt{(n_k+1)|c_k|}\, Q^\times \tilde{\rho}_{\boldsymbol{n}+\boldsymbol{e}_k} \\ & - i\sum_k \sqrt{\frac{n_k}{|c_k|}}\left(c_k Q\tilde{\rho}_{\boldsymbol{n}-\boldsymbol{e}_k} - c_k^*\tilde{\rho}_{\boldsymbol{n}-\boldsymbol{e}_k}Q\right). \end{aligned} \tag{27}$$

The root ADO is the physical reduced density matrix; higher ADOs retain bath memory and system–bath correlations. An occupation vector $\boldsymbol{n} = (n_0, \ldots, n_K)$ records how often each exponential bath mode enters the hierarchy. Truncating at $\sum_k n_k \leq L$ gives

$$N_{\mathrm{ADO}} = \binom{L+K+1}{K+1}, \tag{28}$$

before multiplying by the square of the retained system dimension. Scaling by powers of $|c_k|$ makes neighboring hierarchy tiers numerically comparable and improves conditioning. No Born, Markov, or secular approximation is used within the declared harmonic-bath model. Hierarchy depth, Matsubara count, and retained system states are explicit convergence axes. The declared regime assumes a harmonic Gaussian bath, linear coupling, and Drude spectral density. This follows the nonperturbative reduced-hierarchy construction of Ishizaki and Tanimura (JPSJ 74, 3131 (2005)) and scaled hierarchy stabilization of Shi *et al.* (JCP 130, 084105 (2009)).

Factorized, correlated-equilibrium, and correlated-projected preparations are available. Correlated equilibrium is obtained by solving the sparse stationary truncated HEOM with a unit-trace constraint; a system preparation operator is then applied to every ADO for correlated nonequilibrium initialization. At converged hierarchy and bath expansion, the stationary real-time hierarchy corresponds to the thermal hierarchy generated along imaginary time, as established by Tanimura (JCP 141, 044114 (2014)).

Factorized initialization sets all non-root ADOs to zero and therefore omits initial system–bath correlations. Stationary initialization instead solves $\mathcal{L}_{\mathrm{HEOM}}\boldsymbol{\rho}_{\mathrm{eq}} = 0$ together with $\mathrm{Tr}\,\rho_{\mathbf{0}} = 1$. Applying a system operator A to every tier,

$$\rho_{\boldsymbol{n}}(0) = \frac{A\rho_{\boldsymbol{n}}^{\mathrm{eq}}A^{\dagger}}{\mathrm{Tr}(A\rho_{\mathbf{0}}^{\mathrm{eq}}A^{\dagger})}, \tag{29}$$

creates a correlated nonequilibrium preparation without discarding bath memory. Stationary residual alone is insufficient: root positivity, trace, Hermiticity, ADO activity, and stability against L, K, and system-basis enlargement remain required.

A 100 fs strong-coupling validation used $\lambda = 500$ cm^{-1}, $\gamma^{-1} = 25$ fs, ten retained system states, two Matsubara terms, and hierarchy depth four (35 ADOs). The reorganization/median-gap ratio was 4.47, maximum ADO norm was 0.966, trace and Hermiticity errors were below 5×10^{-16}, and the minimum root-density eigenvalue was -1.3×10^{-16}. Depth, bath-pole, and system-basis variants passed a 0.02 population tolerance; the largest population change was 1.70×10^{-4}.

A separate 10 K validation used eight Matsubara terms, hierarchy depth three, eight retained system states, and 220 ADOs. Depth four, a ninth Matsubara pole, and two extra system states all passed the same tolerance; the largest population change was 3.05×10^{-4}. Low-temperature calculations therefore remain supported, but require additional bath poles and convergence checks.

2.6 Exact non-Gaussian spin bath

`spin_bath_non_gaussian.py` exactly diagonalizes a two-state transfer system coupled to finite spin-1/2 modes,

$$H = H_S + \sum_j \frac{\omega_j}{2}\sigma_j^z + Q\sum_j g_j\sigma_j^x. \tag{30}$$

Thermal product-bath initialization, exact unitary evolution, and exact partial trace yield the reduced density matrix without Gaussian, Born, Markov, or secular assumptions. Specifically,

$$\rho(0) = \rho_S(0) \otimes \frac{e^{-\beta H_B}}{Z_B}, \qquad \rho_S(t) = \mathrm{Tr}_B\left[e^{-iHt}\rho(0)e^{iHt}\right]. \tag{31}$$

Unlike a harmonic bath, spin fluctuations are bounded and possess nonzero higher cumulants, so two-point Gaussian correlation data do not determine the full bath influence. Exact diagonalization retains those statistics and every finite-size recurrence.

Output includes trace distance

$$D(\rho_1, \rho_2) = \frac{1}{2}\|\rho_1 - \rho_2\|_1 \tag{32}$$

for reactant- and product-initialized states. Any interval with increasing D witnesses recovery of state distinguishability and hence information backflow in the Breuer–Laine–Piilo sense (PRL 103, 210401 (2009)).

The reported sum of positive discrete increments is a witness for this chosen state pair, not the optimization over all initial pairs required by the formal BLP measure. This finite non-Gaussian reference model follows exact spin-star treatments such as Semin, Sinayskiy, and Petruccione (PRA 89, 012107 (2014)). Hilbert-space cost is exponential, so the implementation enforces at most ten bath spins; it does not represent an arbitrary macroscopic non-Gaussian environment. Because Hilbert dimension is 2^{N_B+1} and dense diagonalization scales cubically in that dimension, this path is a benchmark for small structured baths rather than a continuum solver. Apparent relaxation must also be distinguished from dephasing followed by finite-size revival.

2.7 Stochastic potential bath

Either diabatic well can receive coordinate-shift, curvature, or enthalpy modulation. Noise can be white or Ornstein–Uhlenbeck colored and can be shared or independent between wells. Colored and white processes begin with their stationary Gaussian marginal distribution. Mode-dependent input amplitudes use length, force-constant, or energy units as appropriate.

For colored modulation, each active surface variable follows the exact discrete OU map

$$R_{n+1} = aR_n + \sigma\sqrt{1-a^2}\,\xi_n, \qquad a = e^{-\Delta t/\tau_c}, qquad\tau_c = 2/\mathrm{FWHM}, \tag{33}$$

giving $\langle R_n^2\rangle = \sigma^2$ and $\langle R(t)R(0)\rangle = \sigma^2 e^{-|t|/\tau_c}$. Shared noise models common-mode environmental motion; independent noise models separate local environments. Common- mode enthalpy shifts largely cancel from the instantaneous diabatic energy gap, whereas anticorrelated or independent shifts broaden it. Coordinate-shift and curvature modes also change forces, so equal numerical σ values across modes do not represent equal physical perturbations.

2.8 Absorbing boundaries

Optional smooth edge masks suppress periodic FFT wraparound. Within an edge layer of width w, each amplitude is multiplied after a full time step by

$$M(x) = \exp\left[-\Gamma_{\mathrm{abs}}\Delta t\left(\frac{d_{\mathrm{edge}}(x)}{w}\right)^p\right]. \tag{34}$$

Absorbed norm represents outgoing flux. Norm-conservation tests apply only when edge absorption is disabled.

The mask is equivalent to a finite-step non-Hermitian edge potential. It prevents wraparound only when outgoing density enters the layer smoothly and is attenuated before reaching the periodic boundary. Too narrow or too strong a layer reflects high-wavenumber components; too weak a layer leaks them through. Production results must therefore be stable under independent variation of layer width, rate, power, and box size. Population lost to the mask cannot be interpreted as reaction yield unless the absorbing region has a physically defined channel meaning.

2.9 Two-coordinate propagation

`multidimensional_transfer.py` propagates $\Psi(x,y,t) = (\psi_1,\psi_2)^T$ with a two-dimensional FFT split operator. Coordinate x is the H-transfer coordinate and y is a promoting or bath mode with independent mass. Shifted harmonic valleys contain reaction-path coupling, and the diabatic coupling is Gaussian in both coordinates. The solver writes physical-unit two-dimensional densities and potential surfaces to HDF5.

For separable Cartesian-like coordinates, the kinetic operator is

$$\hat{T} = -\frac{1}{2m_x}\frac{\partial^2}{\partial x^2} - \frac{1}{2m_y}\frac{\partial^2}{\partial y^2}, \tag{35}$$

so the multidimensional FFT keeps the kinetic exponential diagonal. A promoting mode can alter valley geometry, crossing accessibility, and effective coupling even when it is not itself a dissipative bath. Thus a change relative to one-dimensional dynamics may reflect coherent mode coupling rather than decoherence.

2.10 Three-coordinate propagation and external PES data

`multidimensional_transfer_3d.py` propagates $\Psi(x, y, z, t)$ using a three-dimensional FFT split operator. Coordinate x is H transfer; y and z are independent promoting modes with separate masses and reaction-path couplings. The analytic model or an external Hermitian diabatic PES may supply V_{11}, V_{22}, and complex V_{12}.

The three-coordinate kinetic operator adds $-(2m_z)^{-1}\partial_z^2$. Computational storage and FFT work grow with $2N_xN_yN_z$, illustrating the exponential grid-growth problem for further direct coordinates. Three dimensions therefore provide an explicit coupled-mode reference, not a scalable replacement for reduced-dimensional models, tensor networks, or multiconfiguration methods.

`potential_data.py` defines the external HDF5 schema, rejects missing or incorrect units, validates finite monotone axes and array shapes, performs trilinear interpolation, and converts angstrom/cm^{-1} data to atomic units once. Thus first-principles PES data can enter dynamics without manual unit handling. Electronic- structure generation remains the responsibility of an external code; no dynamics package can infer a first-principles PES from absent electronic-structure data.

Adiabatic energies alone do not uniquely determine a diabatic potential matrix: electronic phases, state tracking, derivative couplings, and the chosen diabatization fix the off-diagonal element and its sign or complex phase. Input surfaces must share one smooth diabatic gauge across the full grid. Interpolation is numerical, not a repair for discontinuous state labels or insufficient electronic-structure sampling. These caveats follow the nonadiabatic-coupling analysis of Baer (Phys. Rep. 358, 75 (2002)) and modern first-principles diabatization discussion by Guan *et al.* (PCCP 23, 24962 (2021)).

3 Units and input boundary

Namelist input uses chemistry-facing units and converts once to atomic units. Output writers convert back to physical units.

Atomic units set $\hbar = e = m_e = 4\pi\epsilon_0 = 1$, removing conversion factors from propagators and bath equations. Conversion occurs once at the input boundary and once at output. This avoids mixing a chemistry-scale time step with an atomic-unit Hamiltonian, a failure that can remain numerically stable while being physically wrong. Density conversion includes the Jacobian: if $x_{\mathrm{bohr}} = cx_{\mathrm{Å}}$, then $\rho_{\mathrm{Å}^{-1}} = c\rho_{\mathrm{bohr}^{-1}}$, preserving $\int \rho(x)\,dx$.

Quantity	Input/output unit	Internal unit
Coordinate, widths	angstrom	bohr
Time	fs	atomic time
Energy and coupling	cm^{-1}	hartree
Force constant	cm^{-1} angstrom^{-2}	hartree bohr^{-2}
Quartic coefficient	cm^{-1} angstrom^{-4}	hartree bohr^{-4}
Rates and FWHM	fs^{-1}	atomic-time^{-1}
Mass	u/amu	electron mass
Temperature	K	inverse hartree through β
Wave number	angstrom^{-1}	bohr^{-1}

4 Primary controls

Control	Meaning
`nx, xmin, xmax`	Periodic real-space grid and physical box.
`dt, nsteps, save_every`	Physical time step, total steps, and output cadence.
`ntraj, seed0`	Number of independent stochastic trajectories and base seed.
`temperature_k, mass_amu`	Thermodynamic temperature and transferred-particle mass.
`gamma, use_colored, kernel, fwhm`	Momentum bath strength and memory kernel.
`damp_reactant, damp_product`	Reactant/product diabatic damping switches.
`pot_model, k1, k2, x1, x2`	Well model, stiffnesses, and minima.
`v1_shift, v2_shift, c4_1, c4_2`	Energy offsets and quartic coefficients.
`v12, sigma, want_coupling`	Diabatic coupling magnitude, width, and switch.
`x0, p0, sigma0`	Initial Gaussian packet center, wave number, and width.
`bath_pot_*`	Stochastic potential mode, amplitude, color, and well selection.
`use_absorber`	Enable smooth edge absorption.
`absorber_width`	Edge-layer width in angstrom.
`absorber_rate`	Maximum mask rate in fs^{-1}.
`absorber_power`	Polynomial smoothness exponent.
`hdf5, out_prefix`	Output format and filename prefix.
`write_initial`	Write exact $t = 0$ snapshot before propagation.
`checkpoint_every`	Atomic, schema-versioned checkpoint cadence in steps.
`restart_from_checkpoint`	Continue from exact trajectory checkpoint.
`parallel_hdf5`	Write trajectory/frame hyperslabs to one MPI-IO HDF5 file.

Input may be supplied as `./qle_1d INPUT.nml`. Running `./qle_1d -help` cannot start a trajectory or overwrite output.

5 Parallel execution and reproducibility

MPI distributes complete trajectories; OpenMP and threaded FFTW accelerate each trajectory. Random streams depend on trajectory number rather than MPI rank. Default output uses independent trajectory files. With Parallel HDF5 enabled, ranks advance synchronized trajectory batches and write nonoverlapping trajectory–frame–grid hyperslabs into one `*.ensemble.h5` file through the HDF5 MPI-IO driver. A two-rank, three-trajectory regression checks layout, physical units, finite values, and norms.

Versioned checkpoint files contain wavefunctions, intrinsic Fortran RNG state, Langevin auxiliary variables, stochastic-potential state, time, step, and output index. Each checkpoint is written to a temporary HDF5 file and installed by an atomic rename. Deterministic FFTW plans permit bitwise comparison between uninterrupted and restarted runs.

Trajectory-number-based random streams make stochastic results invariant to MPI partitioning, while deterministic FFT planning stabilizes restart comparisons. Bitwise restart identity is stronger than conservation alone: it verifies that hidden Markov state—RNG position, OU variables, memory friction, output index, and both wavefunctions—is restored. It does not imply bitwise portability across compilers, FFTW builds, processor instruction sets, or floating-point contraction choices.

6 Output and analysis

HDF5 stores physical coordinates, times, wavefunctions, populations, position expectations, and potential surfaces. Static potential surfaces are stored only in the first frame; stochastic surfaces remain frame-resolved. HDF5 operations fail loudly rather than silently leaving incomplete files. ASCII mode writes snapshot densities, observable time series, and static or frame-resolved PES files.

For pure states, stored coordinate densities are $|\psi_i(x,t)|^2$ and integrate to diabatic populations. For density-matrix solvers they are diagonal coordinate-basis elements, $\rho_i(x,x;t)$, and contain incoherent mixtures as well as coherent wavepackets. Population traces alone discard spatial phase, intersurface coherence, and energy redistribution; mechanism analysis should therefore retain density, coherence, energy, and flux diagnostics when available.

6.1 Wavepacket and phase-space visualization

`ShowDyn.py` reads trajectory files, displays HDF5 trees, plots snapshots, animates densities and surfaces, and plots reactant/product probabilities. For each pure diabatic component it can also evaluate the signed Wigner transform

$$W_i(x,k;t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \psi_i^*(x+y,t)\psi_i(x-y,t)e^{2iky}\,dy, \qquad k = \frac{p}{\hbar}. \tag{36}$$

With x in angstrom, k is reported in angstrom^{-1} and W_i is dimensionless. The normalization is fixed by the marginals

$$\int W_i(x,k;t)\,dk = |\psi_i(x,t)|^2, \qquad \int W_i(x,k;t)\,dx = |\tilde{\psi}_i(k,t)|^2, \qquad \iint W_i\,dx\,dk = P_i. \tag{37}$$

The scalar total nuclear Wigner function is $W_{\mathrm{nuc}} = W_1 + W_2$, which traces over the diabatic electronic state; state-resolved plots remain available. Off-diagonal electronic Wigner coherences are not folded into this scalar quantity.

The implementation constructs the centered autocorrelation at each position and uses an FFT over relative displacement. Correlations extending outside the finite box are set to zero. Because phase-space output scales quadratically in grid size, visualization defaults to a uniform 256-point interpolation that preserves component norm; `-wigner-points` exposes a convergence control. Saved HDF5 files retain the original complex wavefunctions, so an N_x^2 Wigner array need not be stored at every frame. The diverging color normalization is symmetric about zero and fixed for the complete animation; by default its limit is the largest selected-state population divided by π, while `-wigner-limit` supplies an explicit common limit for comparisons between runs. The colorbar ticks and limits therefore do not jump as the wavepacket evolves. Red/blue animations combine time-dependent coordinate density and potential surfaces with positive/negative Wigner lobes. Negative values are a quasiprobability signature of coherent interference, not negative ordinary probability. Momentum range, phase-space grid, box, and snapshot cadence must all be checked before quantitative interpretation.

6.2 Kinetic and mechanism analysis

`analyze_ensemble.py` combines rank/trajectory files or reads the parallel single-file layout, checks common time grids, writes mean and standard-error populations, performs bounded mono- and biexponential fits, and can bootstrap trajectories for empirical rate intervals.

The bounded biexponential model used after detected onset t_0 is

$$P_2(t) = P_2(t_0) + A_s\left[1 - e^{-k_s(t-t_0)}\right] + A_f\left[1 - e^{-k_f(t-t_0)}\right], \quad 0 \le A_s, A_f, \quad A_s + A_f \le 1 - P_2(t_0). \tag{38}$$

Bounds prevent negative channel weights and populations above unity. AIC compares mono- and biexponential descriptive models, while trajectory bootstrap probes sampling uncertainty. Neither procedure identifies a microscopic channel: different coherent, dissipative, or heterogeneous processes can generate nearly identical population curves over a finite window.

`convergence_campaign.py` generates and executes time-step, grid, box, absorber, and ensemble-size variants, then writes CSV and Markdown acceptance tables. `mechanism_attribution.py` runs coupling-off, sub-barrier H, sub-barrier D, and over-barrier controls. It combines classical Wigner above-barrier fractions, prompt transfer, isotope suppression, background, bounded rates, and SEM. A mechanism label is emitted only when all preregistered tests pass; otherwise the result is explicitly inconclusive.

For the initial minimum-uncertainty Gaussian used here, the classical-contamination estimate samples

$$W(x,p) = \frac{1}{\pi}\exp\left[-\frac{(x-x_0)^2}{\sigma_0^2} - \sigma_0^2(p-p_0)^2\right] \tag{39}$$

and counts samples satisfying $V_-(x) + p^2/(2m) \geq V_b$. This does not turn a Wigner distribution into a classical ensemble; it provides a conservative diagnostic of how much initial phase-space weight is already energetically above the selected adiabatic barrier. Coupling-off control tests electronic-surface leakage, H/D comparison tests mass sensitivity, and prompt high-energy transfer calibrates an over-barrier response. No single diagnostic is unique, which is why attribution requires their conjunction.

`heom_convergence.py` runs hierarchy-depth, Matsubara-pole, and retained system-basis variants. It gates population changes together with trace, Hermiticity, and positivity errors.

Convergence has distinct numerical and statistical axes. Time step, grid spacing, box size, basis size, hierarchy depth, and pole count control deterministic bias. Trajectory count controls Monte Carlo uncertainty, which decreases only as $N_{\mathrm{traj}}^{-1/2}$ for independent finite-variance samples. Passing one axis at a time around an unconverged baseline can hide compensating errors; production work should repeat coupled refinements and quote tolerances on observables used for the final claim.

6.3 Reactive flux and energy-resolved transmission

`reactive_flux.py` evaluates state-resolved probability current from saved pure-state trajectories. For diabatic component i,

$$j_i(x,t) = \frac{\hbar}{m}\,\mathrm{Im}\left[\psi_i^*(x,t)\partial_x\psi_i(x,t)\right], \qquad J(x,t) = \sum_i j_i(x,t), \tag{40}$$

where the spatial derivative is evaluated spectrally and all differentiation is performed after converting the stored physical coordinates and wavefunctions to atomic units. The reported current is then converted to probability per femtosecond. For a dividing surface x_s, direct integration of the right-side probability $P_>(t)$ provides the continuity check

$$\frac{dP_>}{dt} = J(x_s,t) - J(x_{\mathrm{seam}},t) - A_>(t), \tag{41}$$

where the periodic-box seam current and optional absorber loss $A_>$ are retained rather than silently attributed to reaction. Signed current is partitioned as $J = J^+ - J^-$, giving forward, backward, and net integrated fluxes and the recrossing fraction F^-/F^+.

An instantaneous phenomenological rate may be formed as

$$k(t) = \frac{J(x_s,t)}{P_R(t)}, \tag{42}$$

but it is accepted as a plateau only when the selected time interval has sufficient reactant population, small relative uncertainty, and negligible fitted slope. The tool also reports multiple-time-origin trajectory

quantities such as

$$C_{fs}^{\mathrm{traj}}(\tau) = \langle J(t) P_{>}(t+\tau) \rangle_t \,, \tag{43}$$

$$C_{ff}^{\mathrm{traj}}(\tau) = \langle J(t) J(t+\tau) \rangle_t \,. \tag{44}$$

These are useful trajectory correlations, but they are not equilibrium Kubo transforms and must not be presented as exact thermal reactive-flux rate theory.

For energy resolution, a windowed temporal transform at incident and product surfaces is constructed,

$$\widetilde{\psi}_i(E, x) = \int dt \, w(t) e^{iEt/\hbar} \psi_i(x, t), \tag{45}$$

and the associated positive-going spectral current gives

$$T(E) = \frac{j_{\mathrm{out}}^{+}(E)}{j_{\mathrm{in}}^{+}(E)}. \tag{46}$$

This ratio is a transmission probability only for a closed, time-independent Hamiltonian with asymptotic surfaces, negligible recrossing, adequate spatial and temporal separation, and a time window containing the full passage. Driving, damping, absorbers between surfaces, or incomplete passage make it an apparent spectral ratio instead. Values appreciably above unity trigger a validity warning; the code does not clip them into a physical-looking answer. Analytic free-particle tests verify the current normalization, flux/side continuity, and unit transmission.

7 Illustrative results

The following plots are generated from executable examples and validation campaigns in this codebase. They demonstrate the range of observables and models available; they are not universal predictions for hydrogen transfer. Every population, rate, and mechanistic conclusion remains conditional on the stated Hamiltonian, bath, initial state, and convergence window.

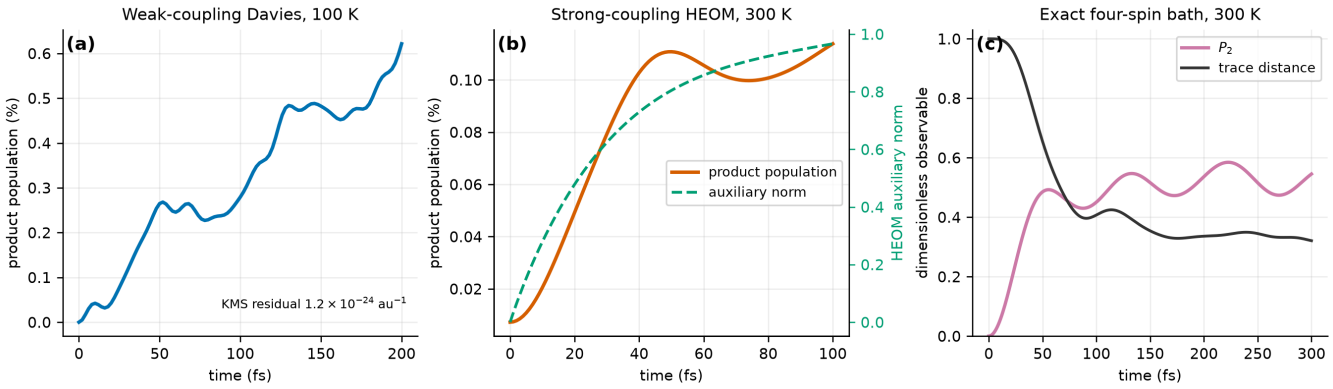


Figure 1: Representative open-system calculations across the implemented model hierarchy. (a) The weak-coupling secular Davies solver produces gradual product transfer while satisfying the KMS stationarity test to numerical precision. (b) A strong-coupling Drude–Lorentz HEOM calculation shows the product population together with the norm carried by auxiliary density operators; their growth is a direct diagnostic that bath memory is active rather than merely parameterized. (c) Exact evolution with four thermal bath spins gives oscillatory transfer and a nonmonotonic trace distance. Local increases in trace distance witness information backflow for the selected state pair, while the finite bath also produces coherent recurrences. Temperatures and time windows differ because these are method-specific validation cases, not a like-for-like physical benchmark.

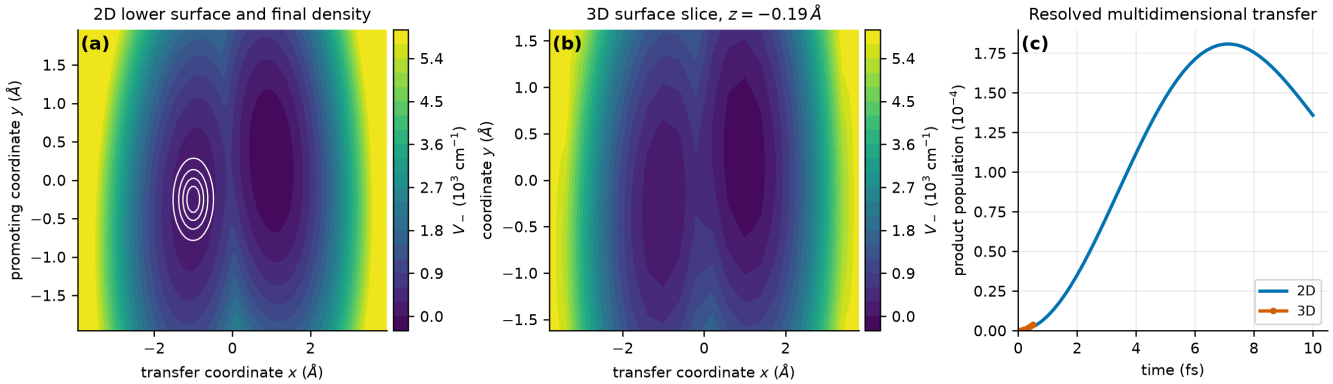


Figure 2: Multidimensional transfer capability. (a) Lower adiabatic surface of the two-coordinate model; white contours show the final total nuclear density and resolve localization in the reactant basin. (b) A fixed-z slice through the three-coordinate lower surface demonstrates the full 3D potential representation. (c) Directly integrated diabatic product populations from the norm-tested 2D and 3D split-operator propagators. The small values are displayed without renormalization; the examples validate propagation and observables rather than claim a converged chemical yield.

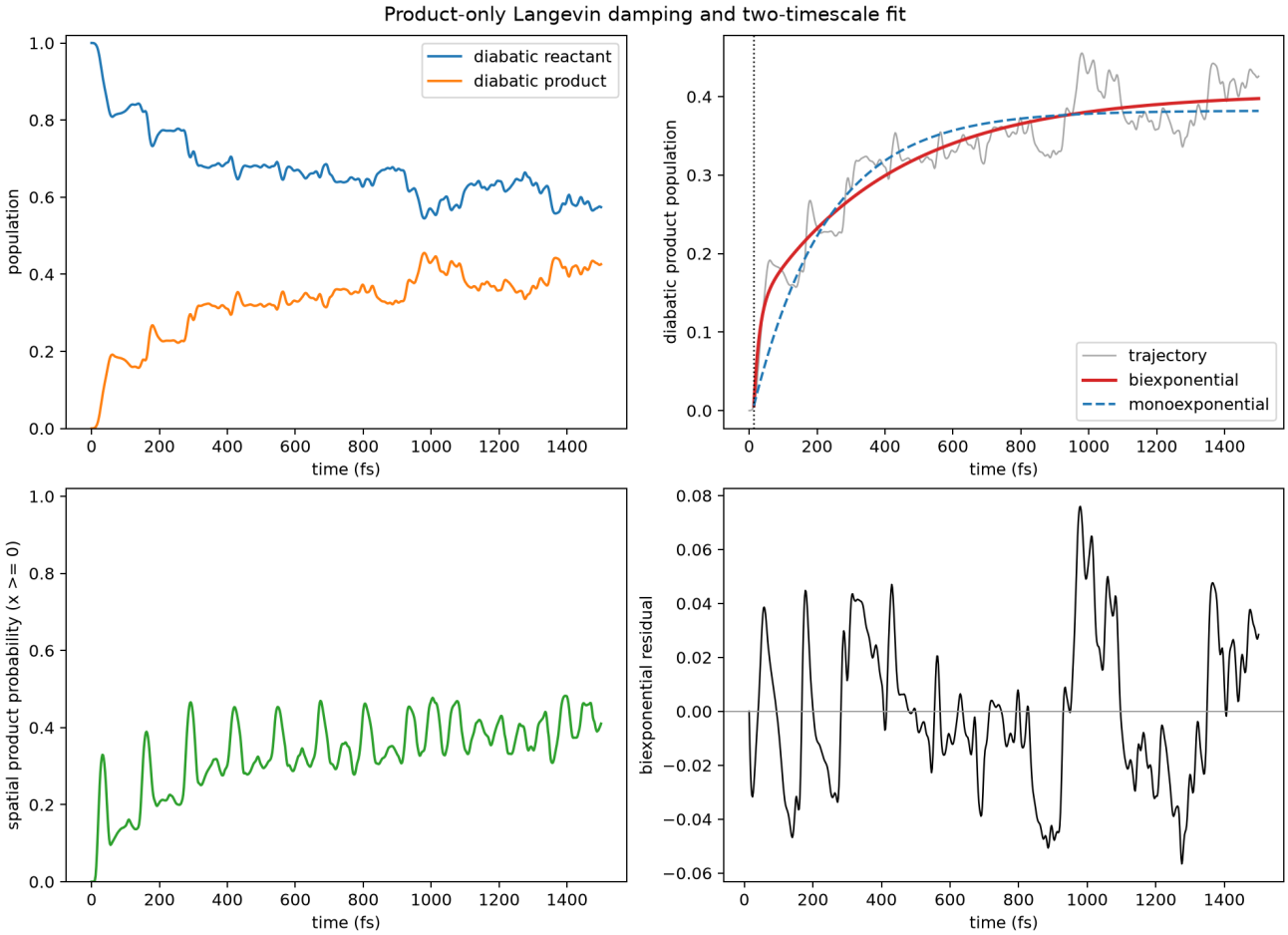


Figure 3: Illustrative 1.5 ps trajectory with Langevin damping restricted to the product diabatic state. The panels show diabatic populations, spatial product probability, bounded mono- and biexponential fits after detected onset, and the biexponential residual. The separated fitted time scales show why the framework is useful for kinetic hypothesis generation. Because this is one stochastic trajectory, the fitted components are descriptive and cannot by themselves establish distinct over-barrier and tunneling mechanisms.

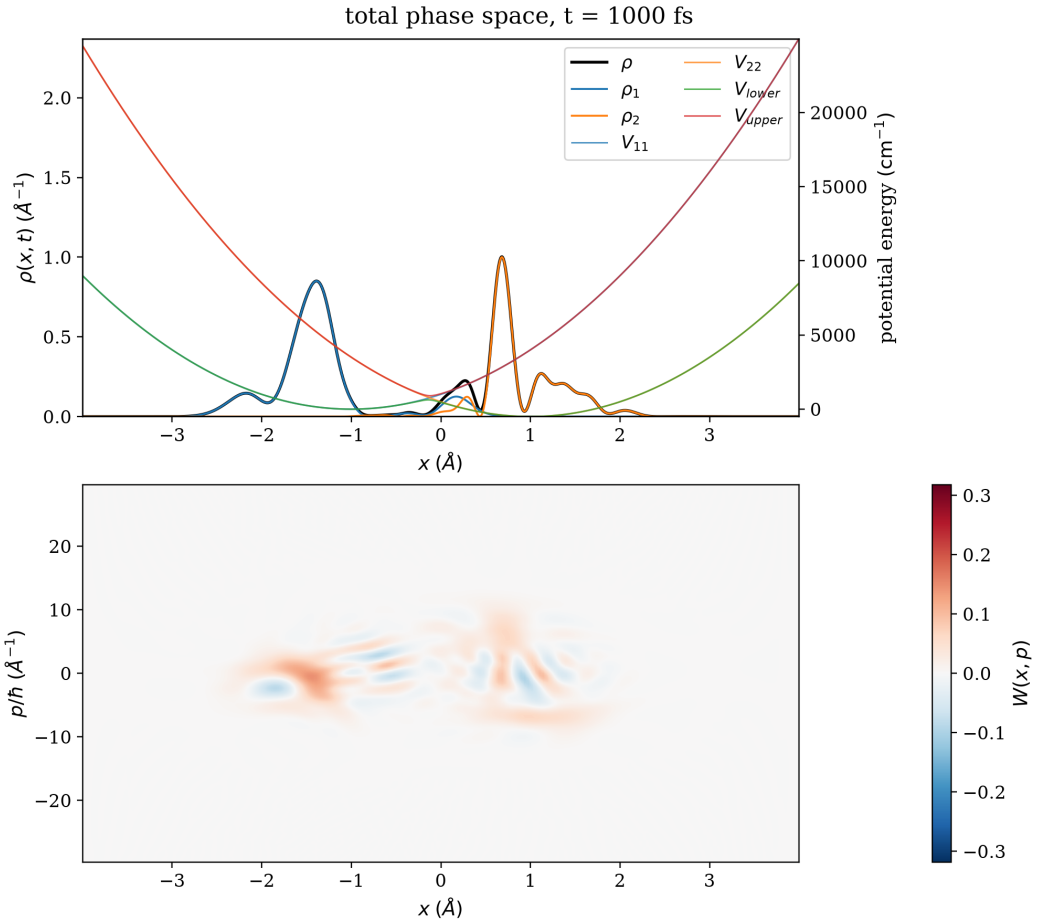


Figure 4: Coordinate-space and phase-space view of the product-damped trajectory at 1.0 ps. The upper panel overlays total and diabatic densities on the diabatic and adiabatic potential surfaces. The lower panel shows the electronic-state-traced nuclear Wigner function versus x and $p/\hbar$. Positive and negative lobes resolve counterpropagating structure and coherent interference hidden by $|\psi(x)|^2$. This single-trajectory quasiprobability is diagnostic rather than a mechanism label; stochastic ensemble claims require averaging consistently sampled trajectories.

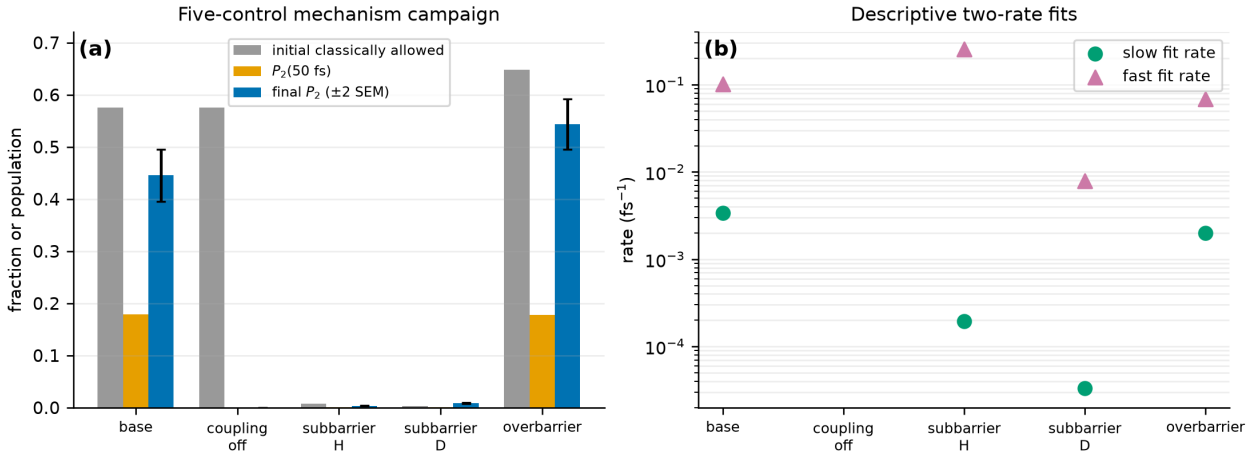


Figure 5: Automated five-control mechanism-attribution campaign, using four trajectories per control. (a) Initial classically allowed Wigner fraction, prompt product population, and final product population with two-SEM error bars. Turning off diabatic coupling eliminates transfer, while the deliberately over-barrier case gives a strong positive response. The sub-barrier isotope controls do not pass the preregistered tunneling gates for this parameter set. (b) Bounded slow and fast fit rates on a logarithmic scale; no rate is reported for coupling-off because transfer is identically zero. The result illustrates an important utility of the code: it can reject an attractive mechanistic interpretation instead of inferring mechanism from a biexponential curve alone.

8 Verification status

Regression coverage includes:

- physical-unit round trips for all supported dimensions;
- zero-damping norm conservation;
- analytic white and colored centroid relaxation;
- fluctuation–dissipation ensemble means and variances;
- product-only damping isolation;
- stationary stochastic-potential initialization;
- coupling and anharmonic-potential switches;
- even/odd FFT frequency ordering and FFTW plan creation;
- smooth absorber norm removal;
- safe command-line help and explicit input paths;
- exact $t = 0$, ASCII, dynamic-PES, and HDF5 output;
- Wigner position and momentum marginals, norm-preserving resampling, negative interference fringes, fixed symmetric color normalization, and end-to-end phase-space rendering;
- analytic plane-wave current, free Gaussian flux/side continuity, recrossing decomposition, and free-particle unit transmission;
- canonical low- and high-temperature limits, exact Gibbs stationarity, and stochastic thermal-pure convergence diagnostics;
- bitwise-identical checkpoint/restart with colored baths;
- two-rank single-file Parallel HDF5 MPI-IO output;
- KMS rate ratios, Gibbs stationarity, density-matrix trace, Hermiticity, and positivity;
- HEOM unitary limit, bath-correlation sum rule, active memory ADOs, strong-coupling invariants, and hierarchy-free departure;
- HEOM hierarchy-depth, Matsubara-pole, and system-basis convergence;
- two-coordinate split-operator norm conservation;
- stationary correlated HEOM residual, nonzero ADO correlations, and correlated-preparation propagation;
- exact finite spin-bath trace, positivity, and information backflow;
- three-coordinate norm conservation and external-PES round trip;
- end-to-end convergence and mechanism-control campaigns;
- HDF5 and non-HDF5 builds under strict compiler warnings.

9 Example two-timescale kinetics

A 1.5 ps product-damped trajectory at 10 K produced a bounded biexponential product-population fit with representative rates

$$k_{\text{fast}} \simeq 6.22 \times 10^{-2}\ \text{fs}^{-1}, \tag{47}$$

$$k_{\text{slow}} \simeq 2.47 \times 10^{-3}\ \text{fs}^{-1}. \tag{48}$$

The fit strongly improved AIC relative to a monoexponential model. Fast and slow components are consistent with prompt/over-barrier and residual tunneling channels, respectively, but this assignment is not unique from one stochastic pure-state trajectory. Publication claims require ensemble statistics and mechanism-control calculations.

A follow-up five-control campaign used four trajectories per control, 1.5 ps duration, and two-SEM decision margins. Coupling-off background was zero and the over-barrier gates passed. The tunneling gates failed: sub-barrier transfer did not exceed estimated classical contamination by the required factor, and D transfer was not suppressed relative to H. Therefore this parameter set supports an over-barrier channel, while the slow biexponential component must remain an empirical timescale rather than a demonstrated tunneling rate.

Rate separation is likewise not mechanism separation. When k_f and k_s approach one another, or when one amplitude is small relative to ensemble noise, biexponential parameters become strongly correlated and poorly identifiable. Stable mechanistic interpretation requires persistence under fit-window changes, bootstrap resampling, longer propagation, convergence refinement, and the independent barrier/coupling/isotope controls above.

10 Completed limitations and method boundaries

Item	Status	Note
Missing $t = 0$ output	Implemented	Exact initial frame now optional and enabled by default.
Periodic boundary return	Implemented	Smooth absorbing layers available; convergence remains user responsibility.
MPI/rank aggregation	Implemented	Ensemble tool reports means, SEM, fits, and bootstrap intervals.
Repeated static HDF5 PES	Implemented	Static PES stored once; stochastic PES remains time resolved.
Single-trajectory rate uncertainty	Implemented	Ensemble bootstrap supported when multiple trajectories exist.
Checkpoint/restart	Implemented	Versioned HDF5 stores all dynamic and random state; atomic replacement and bitwise continuation tested.
Single-file parallel HDF5	Implemented	Parallel-HDF5 build writes one fixed-shape MPI-IO ensemble file.
Quantum FDT	Implemented	Density solver enforces frequency-resolved KMS detailed balance and Gibbs stationarity.
Density-matrix decoherence	Implemented	Complete finite-grid reduced density matrix evolves under a CP Davies generator.
Multidimensional transfer	Implemented	Two- and three-coordinate, two-surface FFT propagation implemented and norm tested.
Automated convergence campaign	Implemented	Time-step, grid, box, absorber, and trajectory-count variants run automatically.
Controlled mechanism assignment	Implemented	Coupling, energy, isotope, prompt-transfer, rate, and SEM gates implemented.

Item	Status	Note
Strong-coupling non-Markovian quantum bath	Implemented	Scaled Drude–Lorentz HEOM retains finite bath memory and system–bath correlations.
HEOM truncation control	Implemented	Hierarchy depth, Matsubara poles, and system-state basis are automatically converged.
Non-Gaussian bath	Implemented	Exact finite spin-bath dynamics implemented; arbitrary macroscopic non-Gaussian baths remain model specific.
Correlated thermal initial state	Implemented	Sparse stationary HEOM hierarchy and correlated system preparations implemented.
Higher than two nuclear coordinates	Implemented	Three explicit nuclear coordinates propagate with a norm-tested FFT solver.
First-principles PES interface	Implemented	Unit-checked HDF5 ingestion and interpolation implemented; electronic-structure generation remains external.
Wigner phase-space diagnostics	Implemented	State-resolved and electronic-state-traced transforms, static plots, and GIF/MP4 animations implemented.
Reactive-flux diagnostics	Implemented	State-resolved current, continuity, recrossing, plateau gates, trajectory correlations, and absorber yields implemented.
Energy-resolved transmission	Implemented	Windowed incident/outgoing spectral flux and validity diagnostics implemented and free-particle tested.
Thermal initial states	Implemented	Exact canonical mixtures and stochastic thermal-pure estimators implemented with convergence diagnostics.

"Definitive" mechanism assignment means definitive only for the declared Hamiltonian, bath model, convergence tolerance, and controls. It is not model-independent experimental proof.

11 Future development roadmap

Reactive-flux diagnostics, energy-resolved transmission, and exact/stochastic thermal initial states were the first three roadmap priorities and are now complete. The following extensions remain future capabilities. Automated isotope campaigns are the next near-term priority.

4. **Automated H/D/T campaigns (near-term priority).** Generate mass-matched controls and bootstrap kinetic isotope effects while testing grid, time-step, duration, ensemble-size, and classical-contamination sensitivity.
5. **Phase-space observables.** Add integrated Wigner negativity, phase-space currents, conditional momentum, coherence lifetimes, and positive Husimi-Q visualization.
6. **Quantum-jump trajectories.** Unravel Davies/Lindblad evolution and verify ensemble agreement with density-matrix dynamics.
7. **Nonsecular Redfield dynamics.** Retain near-degenerate population–coherence coupling, with explicit positivity diagnostics and comparison against Davies and HEOM results.

8. **Structured spectral densities.** Add multiple Drude poles, underdamped Brownian modes, and unit-checked tabulated spectra.
9. **Time-dependent driving.** Add pulsed and periodic bias or coupling modulation for pump–probe, Floquet-assisted, and coherent-control studies.
10. **Richer diabatic Hamiltonians.** Add non-Condon coupling, Morse and multi-barrier surfaces, asymmetric effective masses, and three or more coupled electronic basis states.
11. **Proton-coupled electron transfer.** Couple proton motion to an explicit donor–acceptor or solvent coordinate and electronic-state manifold.
12. **Conical intersections and geometric phase.** Extend complex multidimensional couplings to seam crossing, branching, and Berry-phase tests.
13. **Alternative propagators.** Add Chebyshev, Krylov, DVR, and higher-order split methods as independent accuracy benchmarks.
14. **Uncertainty quantification.** Add global sensitivity, parameter ensembles, bootstrap or Bayesian intervals, and surrogate-assisted fitting.
15. **Embedded provenance.** Store source revision, compiler and dependency versions, build flags, inputs, seeds, convergence evidence, and visualization metadata in HDF5 output.

Here an *electronic state* means one basis channel in the coupled nuclear Hamiltonian. The present one-dimensional wavefunction has two nuclear components and propagates over a 2×2 diabatic matrix formed by V_{11}, V_{22}, and V_{12}. A multistate extension would propagate more nuclear components over an $N \times N$ potential/coupling matrix. Those matrix elements may remain analytic model functions, or they may be generated externally by DFT, multireference, coupled-cluster, or other electronic-structure methods and imported through the unit-checked PES interface. Reliable first-principles use additionally requires state tracking, a consistent electronic phase/gauge, diabatization, interpolation, and uncertainty metadata. This package does not itself perform SCF, DFT, correlated electronic structure, or on-the-fly force calculations. Offline surface/coupling import is therefore the natural first integration step; on-the-fly nonadiabatic dynamics would be a substantially larger framework.

12 Recommended production protocol

1. Run `make test` for the selected compiler and library stack.
2. Perform time-step, grid-spacing, box-size, and absorber convergence.
3. Use several independent trajectories and aggregate with `analyze_ensemble.py`.
4. Monitor total norm, absorbed norm, edge density, and high-frequency FFT content.
5. Compare no-damping, reactant-only, product-only, and both-state damping.
6. Establish mechanism with barrier-energy and coupling controls before labeling fitted rates as over-barrier or tunneling.
7. Archive input, executable revision, dependency versions, seeds, raw data, and analysis outputs.

13 Build and run summary

```
make
make test
make test-parallel
./qle_1d INPUT.nml
python analyze_ensemble.py 'run.traj*.rank*.h5' --output ensemble --bootstrap 500
python ShowDyn.py run.traj000001.rank0.h5 --populations --no-usetex
python quantum_fdt_density.py --initial-preparation gibbs --output quantum_density.h5
python quantum_fdt_density.py --initial-preparation thermal-pure \
  --thermal-samples 4000 --seed 7 --output thermal_pure.h5
python reactive_flux.py run.traj000001.rank0.h5 --surface 0.0 --output reactive_flux.h5
python heom_nonmarkovian.py --output heom_nonmarkovian.h5
```

```
python heom_nonmarkovian.py --initial-preparation correlated-projected
python heom_convergence.py --output heom_convergence
python multidimensional_transfer.py --output transfer_2d.h5
python multidimensional_transfer_3d.py --output transfer_3d.h5
python spin_bath_non_gaussian.py --output spin_bath.h5
python convergence_campaign.py INPUT.nml --executable ./qle_1d
python mechanism_attribution.py INPUT.nml --executable ./qle_1d
python ShowDyn.py run.traj000001.rank0.h5 --wigner --animate \
  --wigner-limit 0.25 --save-animation wigner.mp4
```

HDF5 may be disabled with `make USE_HDF5=0`. FFTW, MPI, OpenMP, and HDF5 prefixes remain configurable through the build environment. True single-file MPI-IO uses `USE_PARALLEL_HDF5=1` with a Parallel-HDF5 installation. MP4, M4V, and MOV animations use H.264/yuv420p with fast-start metadata for QuickTime compatibility when FFmpeg with libx264 is available.